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Minimum dietary diversity and its associated factors among children aged 6–23 months in Mekelle, Northern Ethiopia, following the Tigray war

Nebyu Daniel Amaha^{1*} and Hadush Gebregziabher¹

Abstract

Background Despite the known impact of conflict on child nutrition, limited evidence exists on dietary diversity in urban post-conflict settings. The World Health Organization (WHO) defines Minimum Dietary Diversity (MDD) as the consumption of at least five out of the eight defined food groups in the previous 24 h. This study determined the post-war prevalence of MDD and its associated factors among children aged 6–23 months in Mekelle, northern Ethiopia.

Methods A health-facility-based cross-sectional study of 584 participants was conducted from November to December 2024 in Mekelle using proportional allocation. Sociodemographic and dietary data were collected using a 24-hour dietary recall questionnaire. Associations were assessed using multivariable logistic regression in STATA[®] version 15. Missing values were imputed using single imputation, and sensitivity analyses compared complete-cases ($n=474$) versus imputed cases ($n=584$).

Results Only a quarter (25.2%; 95% CI: 21.6–28.7) of children aged 6–23 months in Mekelle achieved MDD. Grains (92.1%) and legumes (69.3%) were the most commonly consumed food items, while vitamin A-rich foods (26%), other fruits and vegetables (18.2%) and flesh foods (7.2%) were the least. Children aged 18–23 months were 3.2 times more likely (Adjusted Odds Ratio (AOR) = 3.2, $p=0.001$) to achieve MDD than those aged 6–11 months. Children from middle- and high-income households were 6.13-fold (AOR = 6.13, $p<0.001$) and 13.58-fold (AOR = 13.58, $p<0.001$) more likely to meet MDD, respectively. Households with 5–8 members had 3.6-fold higher odds of providing MDD than those with 1–3 members (AOR = 3.6, $p=0.017$). Children whose mothers were aged 26–35 years (AOR = 0.48, $p=0.026$) and 36–42 years (AOR = 0.29, $p=0.017$) had lower odds. Fathers with above-secondary paternal education (AOR = 2.58, $p=0.031$) and merchant occupation (AOR = 3.17, $p=0.001$) were positively associated with MDD when compared to below-secondary and government employment.

Conclusions Post-war dietary diversity in Mekelle remains low and is significantly associated with socioeconomic and demographic factors. The reduced intake of nutrient-dense foods highlights critical gaps in child nutrition.

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Targeted nutrition education for caregivers and interventions promoting dietary diversity are essential in conflict recovery areas.

Keywords Ethiopia, Infant and Young Child Feeding, Malnutrition, Minimum Dietary Diversity, Post-conflict nutrition, Tigray War

Introduction

Armed conflicts exert long-lasting impacts on child nutrition, particularly in low-income settings with fragile food systems. In conflict zones, agricultural production is hindered, food systems are damaged, household incomes plummet, and access to healthcare is restricted. These disruptions exacerbate food insecurity and impair child nutrition, with cascading effects on growth, health, and survival [1, 2]. Evidence from various conflict-affected regions, such as Ethiopia, Côte d'Ivoire, and Uganda, demonstrates links between conflict, reduced dietary diversity, and poor complementary feeding practices [3–7].

Several factors influence dietary diversity in these populations. Pre-conflict determinants such as household wealth, parental education, and market access often continue to impact feeding practices during recovery. However, conflict introduces new barriers, including displacement, loss of livelihoods, and disruption of supply chains, which may alter these determinants [8, 9]. Even educated caregivers may face challenges providing diverse diets if nutrient-dense foods become unavailable or unaffordable due to war-induced inflation and market instability [10].

Post-conflict recovery efforts are vital for restoring child nutrition. In northern Ethiopia, humanitarian interventions and government initiatives have aimed to rehabilitate health infrastructure and restore market functions. Yet, their effectiveness in improving complementary feeding practices remains underexplored. Community-based nutrition programs and social protection schemes are essential to bridge the gap between emergency relief to sustainable recovery [11].

The WHO defines MDD as the proportion of children aged 6–23 months who consume foods from at least five of eight defined food groups within a 24-hour period [12]. As a key Infant and Young Child Feeding (IYCF) indicator, MDD reflects the adequacy of complementary feeding during the critical first 1,000 days of life [13, 14]. Globally, dietary diversity is low, with two-thirds of children failing to meet MDD requirements [14–16]. In Ethiopia, national MDD estimates have declined, from 14.9% in 2016 to 12.6% in 2019 [17, 18].

In northern Ethiopia, the 2020–2022 Tigray War devastated agriculture and healthcare, worsening pre-existing vulnerabilities [11, 19, 20]. Pre-war MDD in Tigray was 26.2% [21], with urban Mekelle at 33.2% based on 2016/2019 Ethiopian Demographic and Health Survey

(EDHS) data [17, 18]. However, the post-conflict effects on urban dietary diversity in are understudied [1]. This study assesses MDD prevalence among children aged 6–23 months in Mekelle two years post-ceasefire and examines sociodemographic and economic factors. These insights can guide recovery efforts and inform interventions in similar conflict-affected regions.

Methodology

Study setting

Mekelle, the capital of Tigray regional state, lies 780 km north of Addis Ababa (Fig. 1) and had an estimated 2024 population of 612,000 [22]. The urban population relies on civil service, trade, and daily labor for livelihoods, with food sourced mainly from markets. Surrounding rural areas cultivate cereals (teff, wheat, barley) and legumes, while common urban foods include injera (fermented flatbread), lentils, beans, and occasional eggs or meat based on affordability. The city is served by five hospitals and eleven health centers [19]. This study, conducted two years post-Tigray war, spanned three of seven sub-cities.

Study design and population

This health-facility-based cross-sectional study was conducted from November to December 2024 in Mekelle. Participants were children aged 6–23 months and their mothers/caregivers attending randomly selected health centers for vaccination. Recruitment used a first-come, first-served (FCFS) approach during vaccination days. Data on sociodemographic characteristics and dietary intake were collected through face-to-face interviews using a standardized questionnaire.

Sampling technique

Three sub-cities were randomly selected from the six with health centers, followed by one health center per sub-city. Children attending these centers on vaccination days were recruited using FCFS approach. The FCFS approach was chosen due to the post-war logistical constraints, potentially introducing selection bias. To enhance representativeness, data were collected across multiple days over four weeks, including morning and afternoon shifts. The sample was proportionally allocated by average monthly child visits. Children whose caregivers did not provide consent, or who had severe chronic illnesses or congenital anomalies, were excluded.

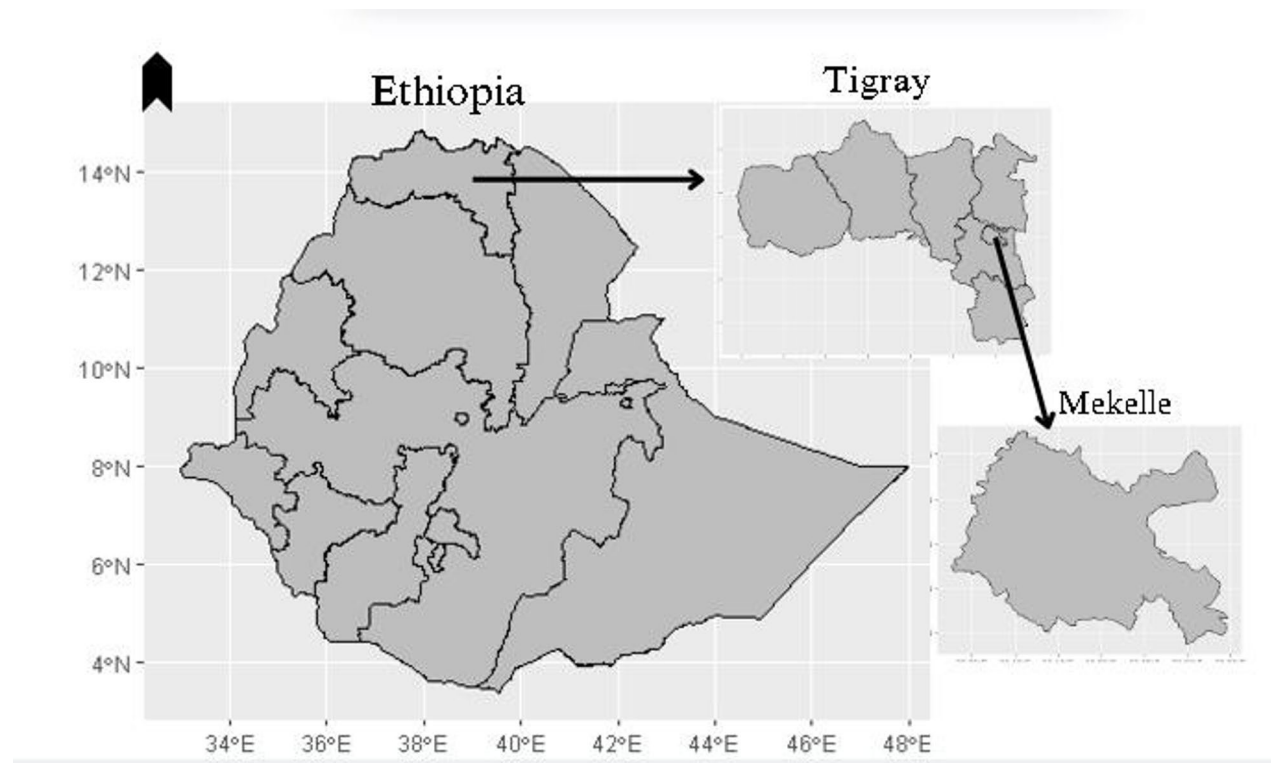


Fig. 1 Geographical location of the study area, Mekelle, the capital of Tigray Regional State, northern Ethiopia

Sample size determination

Sample size was calculated using a double population proportion approach in EpiInfo 7.2 to detect differences in MDD based on previous key factors of maternal education, child age, and wealth [18], with 80% power and a 10% non-response rate. Estimates yielded 330 (maternal education), 584 (child age), and 205 (wealth) (Supplementary Table S2). The largest ($n=584$) was selected to ensure adequate power for robust power.

Data collection

A structured questionnaire, adapted from the 2016 EDHS and FAO tools [17, 23], was used. As these are nationally validated and tested in Ethiopian settings, no separate pilot was conducted [24]. However, the questionnaire was translated into Tigrigna (the local language) and back-translated to English to ensure accuracy and cultural appropriateness. During data collection, completed questionnaires were reviewed daily by supervisors for completeness and consistency. Three BSc Nurses, experienced in surveys, collected data after half-day training session on objectives, modules, and ethics. The principal investigator supervised daily for protocol adherence. Logical checks were conducted during the cleaning phase prior to analysis.

Study variables.

Outcome variable

MDD was defined as consuming of at least five of eight food groups in the last 24 h [12]: (1) breast milk, (2) grains/roots/tubers (3) legumes/nuts, (4) dairy, (5) flesh foods, (6) eggs, (7) vitamin A-rich fruits/vegetables, and (8) other fruits/vegetables [23, 25]. Consumption was binary (1 = consumed, 0 = not consumed), summed as “food-sum,” and categorized as “diverse” (≥ 5 food groups) or “not diverse” (< 5 food groups). Probing questions aided recall (e.g., “Did the child eat any eggs or meat yesterday?”). The mapping of individual food items to the eight WHO food groups is presented in Supplementary Table S1.

Covariates

Variables, selected from prior studies, included: maternal age [26, 27], maternal educational level [10, 28], maternal occupation [29, 30], marital status [31, 32], and place of delivery [33, 34]. Paternal education and occupation [35, 36]. Household income level [10, 37], internet access [27, 38], and household size [27, 29]. Child age [27, 31], child sex [30, 39], birth order and child health status [40]. Child age categories (6–11, 12–17, 18–23 months) followed WHO complementary feeding guidelines [12]. Maternal age groups (17–25, 26–35, and 36–42 years) distinguished between younger mothers, women in peak reproductive years, and those of advanced maternal age [41]. Wealth index was derived by dividing the

household's monthly income in Ethiopian Birr into low, middle, and high terciles. Given the post-conflict context where asset ownership was highly disrupted, household income was used as a socioeconomic proxy instead of a standard wealth index. The terciles were generated using the `xtile` command in Stata, which divides the ranked income variable into three equal-sized groups (33rd and 66th percentiles).

Data analysis

Data were entered into Microsoft Excel, cleaned, and analyzed in Stata version 15. Missingness was ~1% overall.

Table 1 Socio-demographic characteristics of the respondents and their children in Mekelle 2024 ($n=584$)

	Categories	n	%
Child age in months	6–11	236	40.4
	12–17	200	34.2
	18–23	148	25.3
Child sex	Male	267	45.7
	Female	317	54.3
Child birth order	First	201	34.4
	Second	213	36.5
	Third and above	170	29.1
Child had diarrhea last 2 weeks	Yes	156	26.7
	No	428	73.3
Maternal age in years	17–25	168	28.8
	26–35	359	61.5
	36–42	57	9.8
Household members	1–3	224	38.4
	4	205	35.1
	5–8	155	26.5
Religion	Orthodox	517	88.5
	Other	67	11.5
Marital status	Married	536	91.8
	Other	48	8.2
Maternal educational level	Below Secondary	175	29.9
	Secondary	209	35.8
	Above secondary	200	34.2
Maternal occupation	Homemaker	335	57.4
	Government	141	24.1
	Other	108	18.5
Paternal educational level	Below Secondary	144	24.7
	Secondary level	123	21.1
	Above secondary	317	54.3
Paternal occupation	Government	232	39.7
	Merchant	149	25.5
	Other	203	34.8
Household income tercile	Low	195	33.4
	Middle	195	33.4
	High	194	33.2
Maternal internet use	Yes	349	59.8
	No	235	40.2
Place of delivery	Health facility	565	96.7
	Home	19	3.3

The pattern of missing data was non-random, household income (3%), religion (2%), maternal age (2%), egg consumption (2%), husband's education (5%) and occupation (5%). Given that missingness was minimal, single imputation (median for continuous, mode for categorical) approach was deemed appropriate over multiple imputation [42, 43]. Descriptive statistics summarized the data. Binary logistic regression was selected because the outcome variable (MDD) is dichotomous (Met vs. Not Met). Hosmer-Lemeshow assessed model fitness. Multicollinearity checks using the Variance Inflation Factor (VIF) removed birth order ($VIF > 5$; Supplementary Table S3). Multivariable logistic regression identified associations ($p < 0.05$). Sensitivity analysis compared imputed ($n = 584$) and complete-case ($n = 474$) results.

Results

Sociodemographic characteristics of the participants

As shown in Table 1, the study included 584 mother-child pairs. Maternal characteristics showed the most mothers aged 26–35 (61.5%), married (91.8%), with high education (34.2% above-secondary, 35.8% secondary), but, 57.4% homemakers and 59.8% of mothers reported internet use. Paternally, 54.3% had above-secondary education, with 39.7% in government jobs and 25.5% merchants. Children were mostly 6–11 months (40.4%), females (54.3%), with 26.7% recent diarrhea. Nearly all mothers (96.7%) delivered in health facilities, and 88.5% of households identified as Orthodox Christian (Table 1).

Prevalence of minimum dietary diversity

As illustrated in Fig. 2, post-war MDD prevalence was 25.2% (95% CI: 21.6–28.7). Grains were the most commonly consumed food group (92.1%), followed by breast milk (89.6%) and legumes (69.3%). In contrast, nutrient-dense foods like flesh foods (7.2%) and other fruits and vegetables (18.2%) were the least consumed (Fig. 2).

Factors associated with minimum dietary diversity

As presented in Table 2., multivariable logistic regression identified child age, maternal age, household size, paternal education, occupation, and household income as predictors (Table 2.). Middle- and high-income children had higher odds (AOR = 6.13, 95% CI: 2.73–13.78, $p < 0.001$; AOR = 13.58, 95% CI: 5.94–31.07, $p < 0.001$) versus low-income. Older children (12–17 months: AOR = 1.74, 95% CI: 1.02–2.93, $p = 0.039$; 18–23 months: AOR = 3.23, 95% CI: 1.85–5.65, $p < 0.001$) fared better than 6–11 months. Older mothers correlated with lower odds (26–35 years: AOR = 0.48, 95% CI: 0.26–0.92, $p = 0.026$; 36–42 years: AOR = 0.29, 95% CI: 0.1–0.8, $p = 0.017$). Larger households (5–8 members: AOR = 3.6, 95% CI: 1.74–7.44, $p = 0.001$) had higher odds. Paternal above-secondary education (AOR = 2.58, 95% CI: 1.09–6.1, $p = 0.031$)

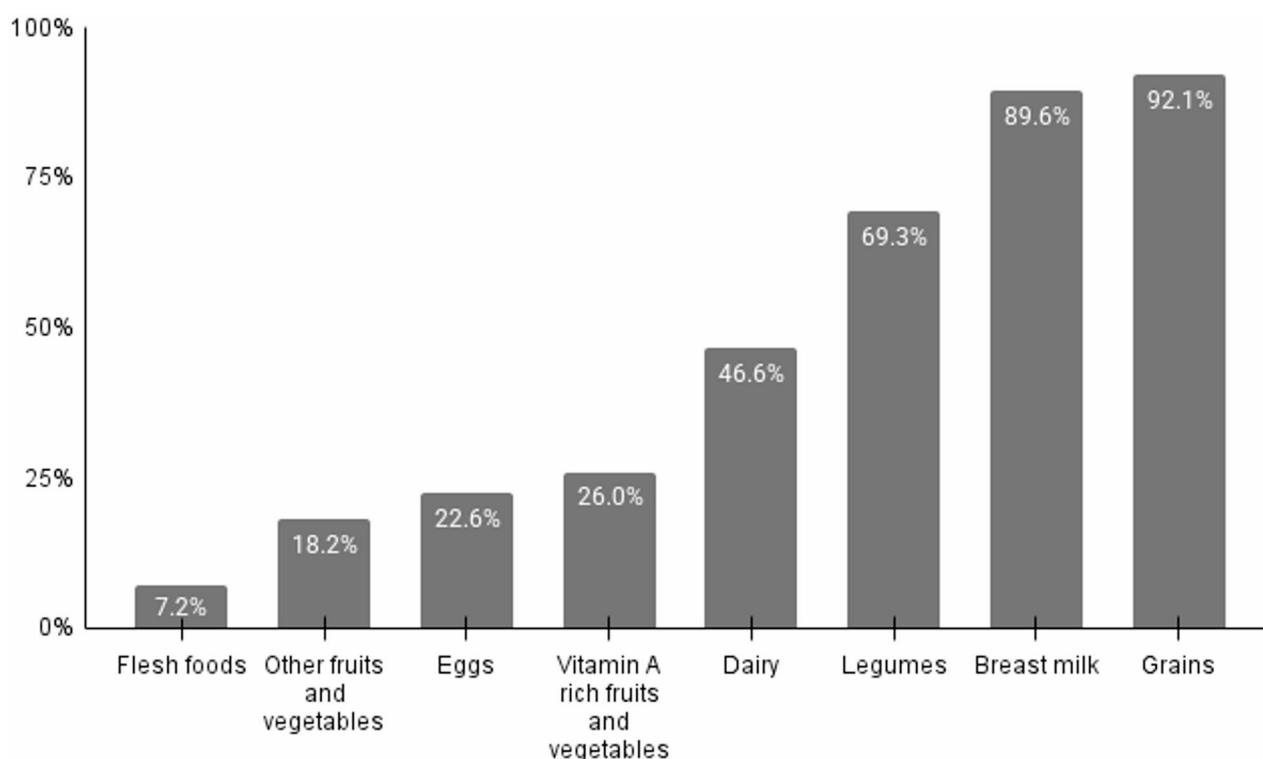


Fig. 2 Percentage of children aged 6–23 months who consumed specific food items the previous day in Mekelle, 2024 ($n=584$)

and merchant occupation ($AOR=3.17$, 95% CI: 1.65–6.11, $p=0.001$) were positive. Model fit was adequate ($\chi^2=11.66$, $p=0.167$). As shown in Supplementary Table S4, sensitivity analysis confirmed robustness for key predictors (e.g., income, older child age, household size, merchant occupation). However, some (e.g., 12–17 months age, 26–35 maternal age) were significant only post-imputation, while maternal secondary education was significant only in complete-cases.

Discussion

This study reveals low dietary diversity among children aged 6–23 months in post-war Mekelle. The increased reliance on grains and legumes, alongside reduced consumption of nutrient-dense foods like flesh foods and vitamin A-rich fruits, highlights the nutritional challenges faced by families in conflict-affected areas. The dominance of grains and legumes in Mekelle mirrors trends in the DRC [44], suggesting a universal conflict-forced transition toward calorie-dense, shelf-stable staples at the expense of perishable, nutrient-rich animal proteins. This shift underscores a common vulnerability across diverse geographic regions that dietary quality remains compromised by the high cost and scarcity of animal-source foods in active and post-conflict zones [5, 45].

The prevalence of MDD in Mekelle was 25.2%. While this figure aligns with findings from rural DRC's (26.0%)

[44] and the Gaza Strip (31.4%) [46], these similarities must be interpreted within their specific socio-political contexts. Both the DRC and Gaza comparisons are meaningful as they represent populations facing protracted instability where traditional food systems have been supplanted by emergency aid and market fragmentation. Locally, Mekelle's MDD (25.2%) was notably higher than the 7.6% reported in rural Amhara [3]. This disparity highlights the urban–rural divide in conflict recovery. As a regional hub, Mekelle benefited from relatively better market access and the centralized distribution of humanitarian fortified foods, whereas rural areas in Amhara faced greater physical isolation and a total collapse of subsistence farming.

Household income and size emerged as predictors of MDD among children in Mekelle. These findings aligns with studies from Indonesia [47], Ethiopia [31, 48], and the DRC [10], showing that household wealth positively influences dietary diversity. Wealthier families can afford a wider variety of nutritious foods, including fresh fruits, vegetables, meat, and dairy products. Children from larger households (5–8 members) were more likely to meet MDD than those from smaller households. This suggests that larger families may benefit from resource pooling and shared labor for food preparation, allowing for more diverse meals despite economic constraints.

Paternal occupation and educational level were also significant predictors of child dietary diversity. Children

Table 2 Multivariable logistic regressions of factors associated with minimum dietary diversity among children aged 6–23 months in Mekelle (n=584)

		Minimum Dietary Diversity		Unadjusted Odds Ratio		Adjusted Odds Ratio	
Categories		Met	Unmet				
		n (%)	n (%)	COR (95%CI)	p-value	AOR (95%CI)	p-value
Child age in months	6–11	45 (30.6)	191 (43.7)	Ref		Ref	
	12–17	51 (34.7)	149 (34.1)	1.45 (0.92 - 2.29)	0.107	1.83 (1.07–3.13)	0.028
	18–23	51 (34.7)	97 (22.2)	2.23 (1.4 - 3.57)	0.001	3.23 (1.81–5.75)	< 0.001
Child sex	Male	62 (42.2)	205 (46.9)	Ref		Ref	
	Female	85 (57.8)	232 (53.1)	1.21 (0.83 - 1.76)	0.319	1.18 (0.75–1.86)	0.483
Child had diarrhea recently	No	114 (77.6)	314 (71.9)	Ref		Ref	
	Yes	33 (22.4)	123 (28.1)	0.73 (0.47 - 1.14)	0.178	0.9 (0.53–1.54)	0.708
Maternal age	17 - 25	37 (25.2)	131 (30.0)	Ref		Ref	
	26 - 35	95 (64.6)	264 (60.4)	1.25 (0.81 - 1.92)	0.308	0.48 (0.26–0.92)	0.026
	36 - 42	15 (10.2)	42 (9.6)	1.18 (0.59 - 2.35)	0.630	0.29 (0.1–0.8)	0.017
Household members	1 – 3	45 (30.6)	179 (41.0)	Ref		Ref	
	4	50 (34.0)	155 (35.5)	1.28 (0.81 - 2.03)	0.284	1.39 (0.76–2.54)	0.285
	5 - 8	52 (35.4)	103 (23.6)	2.01 (1.26 - 3.2)	0.003	3.6 (1.74–7.44)	0.001
Marital status	Married	141 (95.9)	395 (90.4)	Ref		Ref	
	Other	6 (4.1)	42 (9.6)	0.4 (0.17 - 0.96)	0.041	0.77 (0.25–2.42)	0.66
Maternal educational level	Below secondary	28 (19.0)	147 (33.6)	Ref		Ref	
	Secondary	37 (25.2)	172 (39.4)	1.13 (0.66 - 1.93)	0.658	0.78 (0.39–1.59)	0.499
	Above secondary	82 (55.8)	118 (27.0)	3.65 (2.23 - 5.97)	< 0.001	1.42 (0.62–3.27)	0.41
Maternal occupation	Homemaker	62 (42.2)	273 (62.5)	Ref		Ref	
	Government	57 (38.8)	84 (19.2)	2.99 (1.93 - 4.62)	< 0.001	1.79 (0.91–3.51)	0.091
	Other	28 (19.0)	80 (18.3)	1.54 (0.92 - 2.57)	0.097	1.53 (0.81–2.91)	0.191
Paternal educational level	Below secondary	27 (18.4)	117 (26.8)	Ref		Ref	
	Secondary	22 (15.0)	101 (23.1)	0.94 (0.51 - 1.76)	0.856	2.21 (0.98–4.97)	0.056
	Above secondary	98 (66.7)	219 (50.1)	1.94 (1.2 - 3.14)	0.007	2.58 (1.09–6.1)	0.031
Paternal occupation	Government	53 (36.1)	96 (22.0)	Ref		Ref	
	Merchant	59 (40.1)	173 (39.6)	1.62 (1.04 - 2.53)	0.035	3.17 (1.65–6.11)	0.001
	Other	35 (23.8)	168 (38.4)	0.61 (0.38 - 0.98)	0.039	1.95 (0.99–3.84)	0.055
Household income tercile	Low	10 (5.1)	186 (94.9)	Ref		Ref	
	Medium	60 (26.8)	164 (73.2)	6.8 (3.37 - 13.72)	< 0.001	6.13 (2.73–13.78)	< 0.001
	High	77 (46.9)	87 (53.0)	16.46 (8.12 - 33.35)	< 0.001	13.58 (5.94–31.07)	< 0.001
Place of delivery	Health facility	147 (100)	418 (95.7)	-	-	-	-
	Home	0 (0.0)	19 (4.3)	-	-	-	-
Maternal internet use	Yes	114 (77.6)	235 (53.8)	Ref		Ref	
	No	33 (22.4)	202 (46.2)	2.97 (1.93 - 4.57)	< 0.001	1.51 (0.82–2.78)	0.187
Total		147 (25.17)	437 (74.83)				

COR: Crude Odds Ratio; AOR: Adjusted Odds Ratio

of government-employed fathers were less likely to meet MDD than those of merchants, consistent with earlier findings from Ethiopia [35]. This observed association may reflect the limited financial flexibility of fixed salaries compared to business income, which can adapt more readily to economic challenges such as inflation [38]. While this explanation is plausible, it should be noted that this is a hypothesis requiring further investigation through qualitative research on household economic decision-making. Paternal education also showed positive association with MDD. Children whose fathers had higher education levels were more likely to achieve MDD, consistent with studies linking paternal education to

improved child nutrition [35, 39]. Higher education often correlates with better health literacy and higher earning potential, enabling fathers to prioritize and afford diverse foods for their children.

Child and maternal age were positively and negatively associated with MDD, respectively. Children aged 18–23 months were more likely (AOR = 2.8, $p < 0.001$) to achieve MDD than those aged 6–11 months, similar with studies showing that older children are more frequently included in family meals, increasing exposure to a diverse range of foods [27, 49]. This exposure could help develop taste preferences and dietary variety. Maternal age, on the other hand, showed an inverse relationship with MDD.

Children of older mothers (26–35 years and 36–42 years) had lower odds of meeting MDD. This finding is consistent with a study from Nepal, which reported that each additional year of maternal age was associated with a 3% reduction in the likelihood of receiving an adequate diet [27]. Older mothers may face greater household and work-related responsibilities, limiting the time available for child-feeding practices. This underscores the need for targeted nutrition education and support for mothers across different age groups.

Several limitations should be noted. The cross-sectional design prevents causal inferences. Anthropometric measurements were not collected; therefore, the association between Minimally Acceptable Diet (MAD) and child nutritional status (stunting/wasting) could not be assessed. Reliance on 24-hour dietary recall introduces potential recall bias. The FCFS sampling may introduce selection bias, as children arriving earlier might differ systematically from those arriving later. Additionally, unmeasured confounders such as food availability, market prices, and cultural feeding practices were not controlled for. Some secondary associations showed sensitivity to the imputation method suggesting caution when interpreting as they may be influenced by the missing data handling approach. However, the stability of the main socioeconomic and demographic predictors across both analyses strengthens confidence in our primary conclusions regarding the key determinants of dietary diversity in this post-conflict setting. Future research should employ longitudinal designs and incorporate broader dietary assessments to establish causal relationships and explore unmeasured factors such as food availability and market access. Mixed-methods approaches are also recommended to better understand the barriers identified in this study.

Conclusion

Dietary diversity among children aged 6–23 months in post-war Mekelle remained low in the post-war period and is associated with socioeconomic factors such as household income, parental education, and family size. Reduced consumption of nutrient-dense foods like flesh foods and vitamin A-rich fruits, alongside increased reliance on grains and legumes, highlights critical gaps in child nutrition. To address these challenges, interventions should include: (1) targeted nutrition education for caregivers; (2) social protection programs to support low-income households; and (3) initiatives to improve access to diverse, nutrient-rich foods through urban agriculture and market stabilization. Addressing these challenges is essential to mitigate the long-term impacts of conflict on child health and development in Mekelle and similar conflict-affected settings.

Abbreviations

WHO	World Health Organization
MDD	Minimum Dietary Diversity
IYCF	Infant and Young Child Feeding
UNICEF	United Nations Children's Fund
EDHS	Ethiopian Demographic and Health Survey
OR	Odds Ratio
AOR	Adjusted Odds Ratio
VIF	Value Inflation Factor

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s13031-026-00786-1>.

Supplementary Material 1. Table S1. Mapping of 24-Hour Dietary Recall Items to World Health Organization Minimum Dietary Diversity Food Groups. Table S2. Sample size determination using the double population proportion formula. Table S3. Variance Inflation Factor (VIF) of the independent variables before and after removing household members. Table S4. Sensitivity analysis based on the imputed case analysis versus complete case analysis.

Acknowledgements

We sincerely thank all the children and caregivers who participated in this study, and the staff of the health centers in Mekelle for their support during data collection.

Author contributions

NDA: Conceptualized the study, designed the methodology, collected and analyzed the data, and drafted the manuscript. HG: Provided guidance on study design, data analysis, and interpretation of results, and critically reviewed the manuscript. All authors approved the final version of the manuscript.

Funding

No funding was received.

Data availability

The datasets generated and analyzed with this study are available in the DRYAD repository at <https://doi.org/10.5061/dryad.djh9w0w9p>. Moreover, the STATA code used in the analysis is available at <https://github.com/NebyuDanielAmaha/MDD>.

Declarations

Ethics approval and consent to participate

Ethical approval was obtained from the Institutional Review Board (IRB) of the College of Health Sciences, Mekelle University Reference No: MU-IRB 2376/2024, on November 5, 2024. Verbal informed consent was obtained from all participants before data collection. Participants were assured of the confidentiality of their responses and the voluntary nature of their participation.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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Received: 7 September 2025 / Accepted: 10 March 2026

Published online: 15 March 2026

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